Examination : End Semester Examination – January 2025
Name of the Course : B.Tech. (IT & Mathematical Innovations)
Name of the Paper : Data Communication and Networking (DSC-14)
Unique Paper Code : 3122613502
Semester : V
Duration : 3 Hours
Maximum Marks : 90
Instructions: Attempt all questions. Internal choices are given.

1. Answer the following short questions (2 marks each)
   a. Define data communication and list its key components.
   b. Explain the differences between LAN, MAN, and WAN with examples.
   c. What is framing? Explain the concept with an example.
   d. Define and differentiate between flow control and error control.
   e. Define subnet mask and its purpose in IP addressing.
   f. Explain ARP and RARP protocols with use cases.
   g. Explain the difference between connection-oriented and connectionless services.
   h. Define port numbers and their role in the transport layer.
   i. Explain the working of HTTP protocol.
   j. Define cookies and explain their role in web communication.

2. Answer any 5 of the following questions (6 marks each)
   a. Explain the OSI model and its significance in network communication.
   b. Describe the working of CSMA/CD protocol with an example.
   c. Explain Stop-and-Wait ARQ protocol and calculate its efficiency for given parameters.
   d. Differentiate between Distance Vector and Link State Routing algorithms.
   e. Explain IP packet structure with each field description.
   f. Describe TCP three-way handshake process with diagram.
   g. Compare and contrast SMTP, POP3, and IMAP protocols.

3. Answer any 5 of the following questions (8 marks each)
   a. Design basic network topology for a small office setup connecting 10 computers, 1 printer and an internet connection.
   b. In a Go-Back-N ARQ system, if the bit rate is 2 Mbps and 1-way propagation delay is 5 ms, determine the optimal window size for maximum efficiency.
   c. Given dataword 1101 and divisor 1011, compute CRC codeword using modulo-2 Division.
   d. A company is assigned the IP address 192.168.1.0/24. Divide it into 4 equal subnets and specify subnet addresses and range.
   e. If the round-trip time (RTT) is 40 ms and segment size is 1 KB, calculate the throughput using a TCP sliding window of size 4 KB.
   f. Design a simple email transaction showing SMTP commands and responses for sending an email from user A to user B.
   g. Calculate the time required to transfer a 10 MB file using FTP if the link bandwidth is 5 Mbps and 10% time is wasted due to control messages.